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Transient transport equations for high-frequency power flow in heterogeneous cylindrical shells

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Abstract

We use some recent mathematical results obtained for the high-frequency asymptotics of hyperbolic partial differential equations to derive exact transient power flow equations for vibrations of randomly heterogeneous cylindrical shells. The theory shows that the angularly resolved energy densities of an heterogeneous, elastic medium satisfy transient transport equations at higher frequencies. The behaviour of solutions of such equations short of their diffusion limit—if any—is fundamentally different from that of the solution of a diffusion equation, although the latter one is often invoked in the analyses of high-frequency vibrations of elastic structures. A condition by which diffusion equations can be obtained from transport equations is the presence of reflectors or heterogeneities such that scattering mean free paths are short with respect to the characteristic dimensions of the structure. The diffusion limit is reached in this study taking account of scattering by random heterogeneities of the background medium at the scale of the wavelength. This approach fills the gap between transport theory and the diffusion approach in structural dynamics, and clarifies the range of validity of the latter. Our results can be extended to fully coupled dynamic equations for compression, shear and bending of Timoshenko beams or Mindlin plates.

(Some figures in this article are in colour only in the electronic version)

1. Introduction

In their paper which introduced the so-called statistical energy analysis (SEA) of structural vibrations, Lyon and Maidanik [1] showed that the net mean power exchanged by two linearly coupled oscillators is proportional to the difference of their respective mean mechanical energies (the coupling is also assumed conservative). Then they extended this property to the coupling of two multi-modal mechanical systems. Nevertheless, as extensively discussed by

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Woodhouse [2], for instance, this result is not rigorously applicable to general systems having more than two degrees of freedom. This is the reason why SEA considers weakly coupled subsystems in order to recover conditions in which the proportional relation between power flows and mechanical energy differences may be approximately applicable [3].

In a similar vein, it can be shown (Cremer *et al* [4]) that the far field power flow density in an infinitely long, homogeneous Euler–Bernouilli beam in bending is proportional to the gradient of the far field vibratory energy density. This result has been extended to more general structures by Nefske and Sung [5] in order to derive a diffusion equation of heat conduction type for the mechanical energy density, using the local energy balance equation as well. A heat equation analogy of the energy diffusion in a vibrating structure has also been proposed by Rybak [6] directly from the Bethe–Salpeter equation for wave propagation in an infinite plate containing random, delta-correlated heterogeneities under the ladder approximation[1] and steady-state conditions. Such an analogy, referred to as vibrational conductivity, has been invoked further on in the works of Belov *et al* [7] dealing with stiffened plates and Belyaev and Palmov [8] for complex, 'fuzzy' structures under stationary broadband excitations. However, the proportionality of power flows and gradients of energy density, called 'transmission relationship' or 'energetic behaviour law', is not applicable in general cases, and presumed alternative formulations have already been proposed in the literature [9]. This inconsistency has been picked out recently by some researchers [10] who concluded the partial irrelevance of the diffusion equation for vibratory energy flows in linear elastic structures. In the meantime vibrational conductivity analysis of structural dynamics has been the subject of several investigations, but the latter were always limited to simple structures, typically rods, beams and more rarely plates [11]. Uncertainties in the mechanical properties have been systematically ignored in these studies although it is known that their influence on structural vibrations increases as the frequency increases, up to ranges where energy observables are the most relevant. SEA, for instance, accounts implicitly for this feature [12].

The propagation of energy in unbounded randomly perturbed elastic media has been classically dealt with by the Bethe–Salpeter equation in the frame of multiple scattering theory [13]. A radiative transfer equation for the energy intensity can be obtained from that equation under certain circumstances, in particular that the wavelength be small compared to the macroscopic features of the medium and its perturbations be weak. Such an analysis has been proposed in [14] for electrodynamic applications with a statistically homogeneous random potential. Papanicolaou and Burridge [15] applied it to Maxwell's equations for a random medium in the Boltzmann limit of a low density of heterogeneities. As far as elastic media are concerned, such as polycrystals, Weaver [16] has derived a radiative transfer equation in phase-space from a stationary Bethe–Salpeter equation under the ladder approximation, and then obtained diffusion limits. This strategy is similar to the one chosen earlier by Rybak [6], although Rybak considered a much simpler structure and proceeded directly to the diffusion regime without the intermediate radiative transfer step. Later on Turner and Weaver [17] have specialized this approach to the case of an homogeneous thin plate where disorder is introduce by attached random boundary impedance operators. It has also been applied more recently to the case of heterogeneous elastic plates either modelled by Kirchhoff–Love equations [18] (thin plates) or by Navier equations if they are seen as three-dimensional media bounded by two traction-free planes [19]. However only such a leading order—in the perturbations—approximation of the Bethe–Salpeter equation is suitable for analytical considerations. Owing to these limitations, Howe [20] has tried to derive the radiative transfer equation by an alternative approach for a scalar mean field written as the superposition of plane waves.

[1] In this approximation, the K-intensity operator is replaced by the first term of its expansion in correlation groups.

Other authors [21] have used a particular estimator of the energy density, namely the Wigner transform, to derive transfer properties. But it is only recently that a systematic derivation of transport equations for the energy in random electromagnetic, acoustic or elastic media has been proposed [22–25]. The analysis is based on (i) the use of a Wigner transform, whose high-frequency, non-negative limit is the angularly resolved energy density in time and space; and (ii) on the introduction of an explicit scaling of wave propagation patterns. An even more systematic mathematical framework for high-frequency partial differential equations has been derived earlier [26] that precludes the introduction of scales. It originates from considerations on homogenization, but is readily applicable to high-frequency problems in elastodynamics [27, 28]. The main purpose of this paper is to show how this theory can be used for high-frequency vibration analyses of elastic structures, starting with the consideration of some simple ones such as beams or shells. As a general rule, the evolution of the energy density at such frequencies is described by transport equations, instead of the diffusion ones invoked in some of the above-mentioned works. However transport equations can be approximated by diffusion equations under certain circumstances, but the latter are generally not considered or simply ignored in the structural acoustics literature. Our objective is also to clarify the conditions by which this approach shall be valid. These results alleviate a great deal of the inconsistency of the sundry vibrational conductivity approaches developed so far. Several key assumptions they use can also be released: transient transport equations apply to heterogeneous media, possible interferences between propagating modes are not neglected and neither time and/or spatial averages, nor far fields solely are considered. Finally it should be outlined that the theory is corroborated by the trends observed from experimental tests on large-scale, three-dimensional complex structures [29].

This paper is organized as follows. In the next section, we summarize the main mathematical tools used for the analysis of energy propagation in an elastic medium at higher frequencies. In section 3 the theory is applied to the coupled equations for in-plane and bending motions of a cylindrical shell. Transport equations are essentially phase, or propagation direction, dependent. A diffusive regime of heat conduction type can only be obtained in cases where the propagation direction becomes fuzzy, or randomized. This is achieved if we consider, for instance, random perturbations of the background medium which exhibits small deviations about the mean (in order to avoid localization) and long propagation distances. Thus the influence of uncertainties on the mechanical parameters of the shell considered so far is analysed in section 4, while the diffusive regime of heat conduction type is presented briefly in section 5. In section 4 we also present some numerical simulations of the radiative transfer equations obtained for a circular cylindrical shell of finite length, using a direct Monte Carlo method [31] for the approximation of their solutions. At last some conclusions and perspectives are drawn in section 6.

2. Wigner measures as energy density estimates

We summarize in this section (and extend slightly) the basic developments of Lions and Paul [22], Papanicolaou and Ryzhik [23], Gérard *et al* [24] or Guo and Wang [25]. The interested reader shall consult these works for a detailed presentation of the theory and, of course, all proofs of the results given below. Consider a first-order hyperbolic system of the form

$$\boldsymbol{M}(\boldsymbol{x})\partial_t \boldsymbol{u} + \sum_{j=1}^{d} \boldsymbol{D}_j \partial_{x_j} \boldsymbol{u} + \boldsymbol{\Omega}\boldsymbol{u} = \boldsymbol{0} \tag{1}$$

where $\boldsymbol{u}$ is the vector of state variables in $\mathbb{C}^n$, and $\boldsymbol{x} \in \mathbb{R}^d$ runs through the physical space, $\boldsymbol{M}$ is a symmetric, positive definite $n \times n$ real matrix and $\boldsymbol{D}_j$ are symmetric $n \times n$ real matrices independent of position $\boldsymbol{x}$ and time t. At last $\boldsymbol{\Omega}$ is an $n \times n$ real matrix independent of $\boldsymbol{x}$ and t as well. We denote by $\boldsymbol{u}^\epsilon$ the solution of equation (1) for initial conditions oscillating rapidly (in the sense précised in section 2.1) at an arbitrarily small scale ϵ, considering slowly varying mechanical coefficients such that $\boldsymbol{M}$ does not depend on ϵ. We also assume that the sequence $(\boldsymbol{u}^\epsilon)$ is uniformly bounded in $(L^2(\mathbb{R}^d))^n$, where $L^2(\mathbb{R}^d)$ is the set of square integrable functions with respect to the space variable. The setting of equation (1) is very general for wave equations arising from physical problems: it embeds Maxwell's equations for electromagnetic waves, the wave equation for acoustics or the Navier equation for three-dimensional elastodynamics, as shown for instance in [28] or [23]. However it is worth noting here that it is more general than the one proposed in the above works, [23–25] because of the operator $\boldsymbol{\Omega}$. All results below take into account this slight change, which requires a careful examination of all proofs. A further generalization is to consider that this operator also acts as a time convolution product, as well as operators $\boldsymbol{D}_j$. This situation is examined in detail in [30] and allows us to modelize damping with memory effect in an visco-elastic medium or thermal and viscous effects in a poro-elastic medium. It is notably proved that transport equations (12) below are still applicable in this case.

2.1. Energy and power flow estimates

We inject high-frequency energy into the system described by equation (1) through, say, initial conditions of the form

$$\boldsymbol{u}^\epsilon(\boldsymbol{x}, 0) = \boldsymbol{u}_0^\epsilon(\boldsymbol{x}), \tag{2}$$

where the data $\boldsymbol{u}_0^\epsilon$ are bounded in $(L^2(\mathbb{R}^d))^n$ as ϵ goes to 0 and oscillating at most at the frequency $\frac{1}{\epsilon}$. For instance, one can take $\boldsymbol{u}_0^\epsilon(\boldsymbol{x}) = \boldsymbol{u}_0(\boldsymbol{x})\, \mathrm{e}^{\mathrm{i}\boldsymbol{\xi}\cdot\frac{\boldsymbol{x}}{\epsilon}}$, where $\boldsymbol{u}_0$ is square integrable, or $\boldsymbol{u}_0^\epsilon(\boldsymbol{x}) = \epsilon^{-\frac{d}{2}} \boldsymbol{u}_0\big(\frac{\boldsymbol{x}}{\epsilon}\big)$ where $\boldsymbol{u}_0$ is such that there exists a constant $C > 0$ with $\|\nabla \boldsymbol{u}_0\|_{L^2}^2 \leqslant C$, but more general conditions to be fulfilled by the initial data for the following to apply are given in [22, 24]. The mechanical energy density associated with the solution $\boldsymbol{u}^\epsilon$ of equations (1), (2) is defined by

$$\mathcal{E}^\epsilon(\boldsymbol{x}, t) = \tfrac{1}{2}(\boldsymbol{u}^\epsilon, \boldsymbol{u}^\epsilon)_{\boldsymbol{M}} \tag{3}$$

while the densities of power flow along each direction j, $j = 1, 2, \ldots, d$, are

$$\Pi_j^\epsilon(\boldsymbol{x}, t) = \tfrac{1}{2}(\boldsymbol{u}^\epsilon, \boldsymbol{u}^\epsilon)_{\boldsymbol{D}_j} \tag{4}$$

with the notation $(\boldsymbol{u}, \boldsymbol{v})_{\boldsymbol{A}} = (\boldsymbol{A}\boldsymbol{u}, \overline{\boldsymbol{v}})$. A potentially good candidate to study the high-frequency energetic features of the system (1), (2) is the *Wigner transform*, because it has a direct link with the above quantities in the limit $\epsilon \to 0$, as well as some other nice evolution properties given below (see section 2.2). Let us define for two functions $\boldsymbol{u}$ and $\boldsymbol{v}$ lying in a bounded subset of $(L^2(\mathbb{R}^d))^n$:

$$\mathbf{W}^\epsilon(\boldsymbol{u}, \boldsymbol{v})(\boldsymbol{x}, \boldsymbol{k}, t) = \frac{1}{(2\pi)^d} \int_{\mathbb{R}^d} \mathrm{e}^{\mathrm{i}\boldsymbol{k}\cdot\boldsymbol{y}} \boldsymbol{u}\left(\boldsymbol{x} - \frac{1}{2}\epsilon\boldsymbol{y}, t\right) \otimes \overline{\boldsymbol{v}\left(\boldsymbol{x} + \frac{1}{2}\epsilon\boldsymbol{y}, t\right)}\, \mathrm{d}\boldsymbol{y}. \tag{5}$$

Then the $n \times n$ matrix-valued Wigner transform $\mathbf{W}^\epsilon[\boldsymbol{u}^\epsilon] := \mathbf{W}^\epsilon(\boldsymbol{u}^\epsilon, \boldsymbol{u}^\epsilon)$ of $\boldsymbol{u}^\epsilon$ has an Hermitian weak limit (in the sense of temperate distributions) $\mathbf{W}^\epsilon[\boldsymbol{u}^\epsilon] \rightharpoonup \mathbf{W}$ as $\epsilon \to 0$ up to an extracted subsequence which is also a non-negative measure [22, 24], the so-called *Wigner measure* of $\boldsymbol{u}$.

By hyperbolicity, $\boldsymbol{u}^\epsilon$ of (1), (2) has the same oscillatory properties as the initial conditions; thus owing to proposition 1.7 of [24], it can be shown that the energy $\mathcal{E}(\boldsymbol{x}, t)$ and power flow $\Pi_j(\boldsymbol{x}, t)$ densities in the limit $\epsilon \to 0$ are estimated from the Wigner measure by

$$\lim_{\epsilon\to 0} \mathcal{E}^\epsilon(\boldsymbol{x}, t) := \mathcal{E}(\boldsymbol{x}, t) = \frac{1}{2}\int_{\mathbb{R}^d} \boldsymbol{M}(\boldsymbol{x}) : \mathbf{W}(\boldsymbol{x}, \boldsymbol{k}, t)\,\mathrm{d}\boldsymbol{k} \tag{6}$$

and

$$\lim_{\epsilon\to 0} \Pi_j^\epsilon(\boldsymbol{x}, t) := \Pi_j(\boldsymbol{x}, t) = \frac{1}{2}\int_{\mathbb{R}^d} \boldsymbol{D}_j : \mathbf{W}(\boldsymbol{x}, \boldsymbol{k}, t)\,\mathrm{d}\boldsymbol{k}, \qquad j = 1, 2, \ldots, d, \tag{7}$$

respectively, where $\boldsymbol{A} : \boldsymbol{B} = \sum_{j,k} A_{jk} B_{jk}$ stands for the usual tensor scalar product.

2.2. Transport properties of Wigner measures

We now turn to the evolution properties of the Wigner measure associated with the sequence $(\boldsymbol{u}^\epsilon)$ of solutions of the system (1), (2). Define $[\boldsymbol{A}, \boldsymbol{B}] = \boldsymbol{A}\boldsymbol{B} + \boldsymbol{B}^*\boldsymbol{A}^*$, where $\boldsymbol{A}^* = \overline{\boldsymbol{A}}^{\mathrm{T}}$ stands for the adjoint of a matrix, then it satisfies the dispersion equation

$$[\mathrm{i}\boldsymbol{L}, \mathbf{W}] = \boldsymbol{0} \tag{8}$$

where $\boldsymbol{L}$ is the dispersion matrix of the above system defined by

$$\boldsymbol{L}(\boldsymbol{x}, \boldsymbol{k}) = \boldsymbol{M}^{-1}(\boldsymbol{x}) \sum_{j=1}^{d} k_j \boldsymbol{D}_j. \tag{9}$$

It is self-adjoint with respect to the scalar product $(\boldsymbol{u}, \boldsymbol{v})_{\boldsymbol{M}}$: thus its n right eigenvectors form an orthonormal family with respect to this scalar product. The latter and the associated real eigenvalues are denoted by $\boldsymbol{b}_{\alpha_r}(\boldsymbol{x}, \boldsymbol{k})$ and $\lambda_\alpha(\boldsymbol{x}, \boldsymbol{k})$, respectively, for $1 \leqslant \alpha \leqslant M \leqslant n$ and $1 \leqslant r \leqslant R_\alpha$, where R_α is the order of multiplicity of eigenvalue λ_α for a mode α among the M existing ones, such that $\sum_{\alpha=1}^{M} R_\alpha = n$. We assume that R_α does not depend on $(\boldsymbol{x}, \boldsymbol{k})$. From the dispersion equation the Wigner measure may be written

$$\mathbf{W}(\boldsymbol{x}, \boldsymbol{k}, t) = \sum_{\alpha=1}^{M} \sum_{r,s=1}^{R_\alpha} w_\alpha^{rs}(\boldsymbol{x}, \boldsymbol{k}, t) \boldsymbol{b}_{\alpha_r}(\boldsymbol{x}, \boldsymbol{k}) \otimes \boldsymbol{b}_{\alpha_s}(\boldsymbol{x}, \boldsymbol{k}) \tag{10}$$

where the w_α^{rs} are scalar coefficients, hereafter called *specific intensities*. If we define coherence matrices $\boldsymbol{W}_\alpha$, $1 \leqslant \alpha \leqslant M$, by

$$W_{\alpha,rs}(\boldsymbol{x}, \boldsymbol{k}, t) = w_\alpha^{rs}(\boldsymbol{x}, \boldsymbol{k}, t), \qquad 1 \leqslant r, s \leqslant R_\alpha \tag{11}$$

then they satisfy the following transport equations:

$$\partial_t \boldsymbol{W}_\alpha + \{\lambda_\alpha, \boldsymbol{W}_\alpha\} + [\boldsymbol{W}_\alpha, \boldsymbol{Q}_\alpha] = \boldsymbol{0}. \tag{12}$$

$\{\ \}$ stands for the usual Poisson's bracket defined by $\{f, g\} = \boldsymbol{\nabla}_{\boldsymbol{k}} f \cdot \boldsymbol{\nabla}_{\boldsymbol{x}} g - \boldsymbol{\nabla}_{\boldsymbol{x}} f \cdot \boldsymbol{\nabla}_{\boldsymbol{k}} g$ and $\boldsymbol{Q}_\alpha = \boldsymbol{\Omega}_\alpha + \boldsymbol{N}_\alpha$, $1 \leqslant \alpha \leqslant M$, where $\Omega_{\alpha,rs} = \left(\boldsymbol{b}_{\alpha_r}, \boldsymbol{b}_{\alpha_s}\right)_{\boldsymbol{\Omega}}$ and the $\boldsymbol{N}_\alpha$ are skew-symmetric matrices given by

$$N_{\alpha,rs} = \sum_{j=1}^{d} \left(\partial_{x_j} \boldsymbol{b}_{\alpha_r}, \boldsymbol{b}_{\alpha_s}\right)_{\boldsymbol{D}_j} - \left(\boldsymbol{\nabla}_{\boldsymbol{x}}\lambda_\alpha \cdot \boldsymbol{\nabla}_{\boldsymbol{k}} \boldsymbol{b}_{\alpha_r}, \boldsymbol{b}_{\alpha_s}\right)_{\boldsymbol{M}} - \frac{1}{2}\boldsymbol{\nabla}_{\boldsymbol{k}} \cdot \boldsymbol{\nabla}_{\boldsymbol{x}} \lambda_\alpha \delta_{rs}, \qquad 1 \leqslant r, s \leqslant R_\alpha. \tag{13}$$

These rotation matrices arise from the curvature of rays in heterogeneous media, leading to rotation of the polarization vector around a ray as waves propagate [23]; we will see in the following section dealing with cylindrical shells that they vanish for such structures. Of course, if $R_\alpha = 1$ the right-hand side in equation (12) always vanishes, and the latter becomes

$$\partial_t w_\alpha + \{\lambda_\alpha, w_\alpha\} = 0 \tag{14}$$

dropping subscripts r and s.

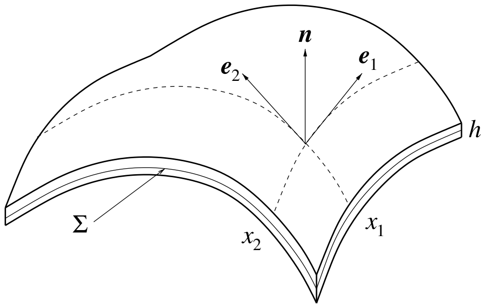

Figure 1. An elastic shell structure occupying the domain $\Sigma \times \left(-\frac{h}{2}, \frac{h}{2}\right)$ of $\mathbb{R}^3$. Σ is the middle surface of the shell, h is its thickness and $(\boldsymbol{e}_1, \boldsymbol{e}_2)$ is its natural tangent basis at the point $\boldsymbol{x} = (x_1, x_2)$. $\boldsymbol{n}$ is the unit normal outward to the middle surface.

3. Transport of energy in a cylindrical shell

The results of the previous section are now applied to a thick-curved shell $\Sigma \times \left(-\frac{h}{2}, \frac{h}{2}\right)$ of which vibrations are described by the model of Naghdi and Cooper [32]. The (regular) middle surface Σ is parametrized by its curvilinear coordinates (x_1, x_2) and the associated natural tangent basis is denoted by $(\boldsymbol{e}_1, \boldsymbol{e}_2)$ such that $\boldsymbol{e}_1 = \partial_{x_1}\boldsymbol{x}$ and $\boldsymbol{e}_2 = \partial_{x_2}\boldsymbol{x}$. Thus $d = 2$ with the notations of the previous section. Tangent motions of the middle surface are denoted by $\boldsymbol{u}_\Sigma$, v is the normal displacement and $\boldsymbol{\theta}$ is the vector of changes of slope of the normal $\boldsymbol{n} = \frac{e_1 \times e_2}{|e_1 \times e_2|}$ to Σ (see figure 1). The second-ordered, normal curvature tensor $\mathbb{B}$ of the shell is

$$\mathbb{B} = \sum_{\alpha,\beta=1}^{2} B_{\alpha\beta}\boldsymbol{e}_\alpha \otimes \boldsymbol{e}_\beta, \qquad B_{\alpha\beta} = -\boldsymbol{e}_\alpha \cdot \partial_\beta \boldsymbol{n} = \boldsymbol{n} \cdot \partial_\beta \boldsymbol{e}_\alpha$$

and it is symmetric. Its inverse trace $(\mathrm{Tr}\,\mathbb{B})^{-1}$, the mean radius of the shell at each point of Σ, is assumed to be much larger than the thickness h, which in turn is significantly lower than a typical wavelength of the excitations.

3.1. Basic equations

Let $\varrho(\boldsymbol{x})$ be the mass per unit surface of the isotropic material constituting the shell and let C be its fourth-order elasticity tensor, which then depends on two parameters. Let $E(\boldsymbol{x})$ be its Young's modulus, $G(\boldsymbol{x})$ its shear modulus such that $G = \frac{E}{2(1+\nu)}$, $\kappa < 1$ its shear reduction coefficient and ν its Poisson's coefficient. Denoting by $\boldsymbol{I}_j$ the $j \times j$ real identity matrix for an integer j, the strain–stress relationship (Hooke's law) is written as

$$\mathrm{C} : \boldsymbol{\epsilon} = \frac{\nu E}{1-\nu^2}(\mathrm{Tr}\,\boldsymbol{\epsilon})\boldsymbol{I}_d + \frac{E}{1+\nu}\boldsymbol{\epsilon}$$

for the second-order linearized strain tensor $\boldsymbol{\epsilon}$, and the membrane forces tensor $\mathbb{N}$, shear forces $\boldsymbol{T}$ and bending moments tensor $\mathbb{M}$ are given by

$$\begin{aligned} \mathbb{N} &= h\mathrm{C} : (\nabla_\Sigma \otimes_s \boldsymbol{u}_\Sigma - v\mathbb{B}) \\ \boldsymbol{T} &= \kappa G h\,(\nabla_\Sigma v - \boldsymbol{\theta} - \mathbb{B}\boldsymbol{u}_\Sigma) \\ \mathbb{M} &= \frac{h^3}{12}\mathrm{C} : \nabla_\Sigma \otimes_s \boldsymbol{\theta}, \end{aligned} \tag{15}$$

where $\otimes_s$ is the symmetrized tensor product and $\nabla_\Sigma = (\partial_1, \partial_2)$ is the gradient on the middle surface. The equations of motion are

$$\varrho\partial_t^2\boldsymbol{u}_\Sigma = \mathbf{Div}_\Sigma\mathbb{N} + \mathbb{B}\boldsymbol{T} \qquad \varrho\partial_t^2 v = \mathrm{div}_\Sigma\boldsymbol{T} + \mathbb{B} : \mathbb{N} \qquad \frac{\varrho h^2}{12}\partial_t^2\boldsymbol{\theta} = \mathbf{Div}_\Sigma\mathbb{M} + \boldsymbol{T}. \tag{16}$$

Let $\boldsymbol{u} = (\partial_t\boldsymbol{u}_\Sigma, \partial_t v, \partial_t\boldsymbol{\theta}, \boldsymbol{N}, \boldsymbol{T}, \boldsymbol{M})$, where $\boldsymbol{N}$ and $\boldsymbol{M}$ are the three-component vector counterparts of $\mathbb{N}$ and $\mathbb{M}$, respectively, which are symmetric. For the model of Naghdi and Cooper considered in this analysis, the normal curvature tensor is $\mathbb{B} = -\frac{1}{R}\boldsymbol{e}_2 \otimes \boldsymbol{e}_2$ if R is the radius of the circular cylindrical shell. Then $n = 13$ and equations (15), (16) are equivalent to the system (1) with

$$\boldsymbol{M}(\boldsymbol{x}) = \begin{bmatrix} \varrho\boldsymbol{I}_2 & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \varrho & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & \frac{\varrho I}{h}\boldsymbol{I}_2 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & (C\boldsymbol{B})^{-1} & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & \frac{1}{Gh} & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{\kappa Gh}\boldsymbol{I}_2 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & (D\boldsymbol{B})^{-1} & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 & \frac{1}{GI} \end{bmatrix},$$

where

$$\boldsymbol{B} = \begin{bmatrix} 1 & \nu \\ \nu & 1 \end{bmatrix}, \qquad C = \frac{Eh}{1-\nu^2}, \qquad D = \frac{EI}{1-\nu^2}, \qquad I = \frac{h^3}{12}.$$

Matrix $\boldsymbol{\Omega}$ is skew-symmetric and given by

$$-\boldsymbol{\Omega} = \begin{bmatrix} 0 & 0 & 0 & 0 & 0 & \mathbb{B} & 0 & 0 \\ & 0 & 0 & \boldsymbol{d}^{\mathrm{T}} & 0 & 0 & 0 & 0 \\ & & 0 & 0 & 0 & \boldsymbol{I}_2 & 0 & 0 \\ & & & 0 & 0 & 0 & 0 & 0 \\ & & & & 0 & 0 & 0 & 0 \\ & & & & & 0 & 0 & 0 \\ & & & & & & 0 & 0 \\ & & & & & & & 0 \end{bmatrix}$$

where $\boldsymbol{d} = \left(0, -\frac{1}{R}\right)$. At last expressions for matrices $\boldsymbol{D}_1$ and $\boldsymbol{D}_2$ are not reproduced here but they may be easily deduced from the previous definition of the state variable $\boldsymbol{u}$.

3.2. *Energy propagation properties*

The dispersion matrix $\boldsymbol{L}(\boldsymbol{x}, \boldsymbol{k})$ of a cylindrical shell is

$$-\boldsymbol{L} = \begin{bmatrix} 0 & 0 & 0 & \frac{1}{\varrho}\boldsymbol{K}(\boldsymbol{k}) & \frac{1}{\varrho}\boldsymbol{m}(\boldsymbol{k}) & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & \frac{1}{\varrho}\boldsymbol{k}^{\mathrm{T}} & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 & \frac{h}{\varrho I}\boldsymbol{K}(\boldsymbol{k}) & \frac{h}{\varrho I}\boldsymbol{m}(\boldsymbol{k}) \\ C\boldsymbol{B}\boldsymbol{K}(\boldsymbol{k}) & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ Gh\boldsymbol{m}(\boldsymbol{k})^{\mathrm{T}} & 0 & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & \kappa Gh\boldsymbol{k} & 0 & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & D\boldsymbol{B}\boldsymbol{K}(\boldsymbol{k}) & 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & GI\boldsymbol{m}(\boldsymbol{k})^{\mathrm{T}} & 0 & 0 & 0 & 0 & 0 \end{bmatrix} \tag{17}$$

where

$$\boldsymbol{k} = (k_1, k_2), \qquad \boldsymbol{K}(\boldsymbol{k}) = \begin{bmatrix} k_1 & 0 \\ 0 & k_2 \end{bmatrix}, \qquad \boldsymbol{m}(\boldsymbol{k}) = (k_2, k_1).$$

Let $|\boldsymbol{k}| = \sqrt{k_1^2 + k_2^2}$, $\hat{\boldsymbol{k}} = \frac{1}{|\boldsymbol{k}|}(k_1, k_2)$ and $\hat{\boldsymbol{l}} = \frac{1}{|\boldsymbol{k}|}(-k_2, k_1)$. Then its eigenvalues on the set $\mathcal{A} = \mathbb{R}^2 \times \mathbb{R}^2 \setminus \{(\boldsymbol{x}, \boldsymbol{k}); \boldsymbol{k} = \boldsymbol{0}\}$ are

$$\begin{aligned} \lambda_0 &= 0 && \text{with multiplicity 3,} \\ \lambda_{\mathrm{T}}^{\pm} &= \pm c_{\mathrm{T}}|\boldsymbol{k}| && \text{each with multiplicity 1,} \\ \lambda_{\mathrm{S}}^{\pm} &= \pm c_{\mathrm{S}}|\boldsymbol{k}| && \text{each with multiplicity 2,} \\ \lambda_{\mathrm{P}}^{\pm} &= \pm c_{\mathrm{P}}|\boldsymbol{k}| && \text{each with multiplicity 2,} \end{aligned} \tag{18}$$

with $c_{\mathrm{S}}^2 = \frac{Gh}{\varrho}$, $c_{\mathrm{T}}^2 = \kappa c_{\mathrm{S}}^2$ and $c_{\mathrm{P}}^2 = \frac{C}{\varrho}$, and the associated eigenvectors on $\mathcal{A}$ are

$$\begin{aligned} \boldsymbol{b}_{1,2} &= \left(0, 0, 0, 0, 0, \pm\sqrt{\frac{\kappa Gh}{2}}\hat{\boldsymbol{l}}, \sqrt{\frac{EI}{2}}\boldsymbol{K}(\hat{\boldsymbol{l}})\hat{\boldsymbol{l}}, \frac{1}{2}\sqrt{\frac{EI}{2}}\boldsymbol{m}(\hat{\boldsymbol{l}})\cdot\hat{\boldsymbol{l}}\right) \\ \boldsymbol{b}_3 &= \left(0, 0, 0, \sqrt{Eh}\boldsymbol{K}(\hat{\boldsymbol{l}})\hat{\boldsymbol{l}}, \frac{1}{2}\sqrt{Eh}\boldsymbol{m}(\hat{\boldsymbol{l}})\cdot\hat{\boldsymbol{l}}, 0, 0, 0\right) \\ \boldsymbol{b}_{\mathrm{T}}^{\pm} &= \left(0, \frac{1}{\sqrt{2\varrho}}, 0, 0, 0, \mp\sqrt{\frac{\kappa Gh}{2}}\hat{\boldsymbol{k}}, 0, 0\right) \\ \boldsymbol{b}_{\mathrm{Sb}}^{\pm} &= \left(0, 0, \mp\sqrt{\frac{h}{2\varrho I}}\hat{\boldsymbol{l}}, 0, 0, 0, \sqrt{(1-\nu)D}\boldsymbol{K}(\hat{\boldsymbol{k}})\hat{\boldsymbol{l}}, \frac{1}{2}\sqrt{2GI}\boldsymbol{m}(\hat{\boldsymbol{k}})\cdot\hat{\boldsymbol{l}}\right) \\ \boldsymbol{b}_{\mathrm{Sn}}^{\pm} &= \left(\pm\frac{1}{\sqrt{2\varrho}}\hat{\boldsymbol{l}}, 0, 0, \sqrt{(1-\nu)C}\boldsymbol{K}(\hat{\boldsymbol{k}})\hat{\boldsymbol{l}}, \frac{1}{2}\sqrt{2Gh}\boldsymbol{m}(\hat{\boldsymbol{k}})\cdot\hat{\boldsymbol{l}}, 0, 0, 0\right) \\ \boldsymbol{b}_{\mathrm{Pb}}^{\pm} &= \left(0, 0, \mp\sqrt{\frac{h}{2\varrho I}}\hat{\boldsymbol{k}}, 0, 0, 0, \sqrt{\frac{D}{2}}\boldsymbol{B}\boldsymbol{K}(\hat{\boldsymbol{k}})\hat{\boldsymbol{k}}, \frac{1}{2}\sqrt{(1-\nu)GI}\boldsymbol{m}(\hat{\boldsymbol{k}})\cdot\hat{\boldsymbol{k}}\right) \\ \boldsymbol{b}_{\mathrm{Pn}}^{\pm} &= \left(\pm\frac{1}{\sqrt{2\varrho}}\hat{\boldsymbol{k}}, 0, 0, \sqrt{\frac{C}{2}}\boldsymbol{B}\boldsymbol{K}(\hat{\boldsymbol{k}})\hat{\boldsymbol{k}}, \frac{1}{2}\sqrt{(1-\nu)Gh}\boldsymbol{m}(\hat{\boldsymbol{k}})\cdot\hat{\boldsymbol{k}}, 0, 0, 0\right). \end{aligned} \tag{19}$$

Given the projection (10) of the Wigner measure (on $\mathcal{A}$), let $\boldsymbol{W}_{\mathrm{P}}^{\pm}(\boldsymbol{x}, \boldsymbol{k}, t)$ be the 2×2 coherence matrices of specific intensities for bending and in-plane waves corresponding to eigenvectors $\boldsymbol{b}_{\mathrm{Pb}}^{\pm}(\boldsymbol{x}, \boldsymbol{k})$ and $\boldsymbol{b}_{\mathrm{Pn}}^{\pm}(\boldsymbol{x}, \boldsymbol{k})$. Similarly, let $\boldsymbol{W}_{\mathrm{S}}^{\pm}(\boldsymbol{x}, \boldsymbol{k}, t)$ be the 2×2 coherence matrices of specific intensities corresponding to eigenvectors $\boldsymbol{b}_{\mathrm{Sb}}^{\pm}(\boldsymbol{x}, \boldsymbol{k})$ and $\boldsymbol{b}_{\mathrm{Sn}}^{\pm}(\boldsymbol{x}, \boldsymbol{k})$. Then they satisfy the following transport equations:

$$\partial_t \boldsymbol{W}_{\alpha}^{\pm} + \left\{\lambda_{\alpha}^{\pm}, \boldsymbol{W}_{\alpha}^{\pm}\right\} = \boldsymbol{0} \tag{20}$$

since rotation matrices $\boldsymbol{N}_{\mathrm{P}}^{\pm}(\boldsymbol{x}, \boldsymbol{k})$ and $\boldsymbol{N}_{\mathrm{S}}^{\pm}(\boldsymbol{x}, \boldsymbol{k})$ as given by equation (13) are null for the present case, as well as $\boldsymbol{\Omega}_{\mathrm{P}}^{\pm}$ and $\boldsymbol{\Omega}_{\mathrm{S}}^{\pm}$. As regards shear waves, specific intensities $w_{\mathrm{T}}^{\pm}(\boldsymbol{x}, \boldsymbol{k}, t)$ satisfy the equations

$$\partial_t w_{\mathrm{T}}^{\pm} + \left\{\lambda_{\mathrm{T}}^{\pm}, w_{\mathrm{T}}^{\pm}\right\} = 0, \tag{21}$$

and they are uncoupled from the bending modes. At last the 3×3 coherence matrix $\boldsymbol{W}_0(\boldsymbol{x}, \boldsymbol{k}, t)$ satisfies

$$\partial_t \boldsymbol{W}_0 = \boldsymbol{0} \tag{22}$$

since here, again, $\boldsymbol{N}_0 = \boldsymbol{\Omega}_0 = \boldsymbol{0}$; thus $\boldsymbol{W}_0(\boldsymbol{x}, \boldsymbol{k}, t) = \boldsymbol{0}$ if it is initially null. We observe that $w_{\mathrm{T}}^-(\boldsymbol{x}, \boldsymbol{k}, t) = w_{\mathrm{T}}^+(\boldsymbol{x}, -\boldsymbol{k}, t)$ and $\boldsymbol{W}_\alpha^-(\boldsymbol{x}, \boldsymbol{k}, t) = \boldsymbol{W}_\alpha^+(\boldsymbol{x}, -\boldsymbol{k}, t)$ for $\alpha =$ S or P, so that the energy density is estimated by

$$\mathcal{E}(\boldsymbol{x}, t) = \int_{\mathbb{R}^2} w_{\mathrm{T}}^+(\boldsymbol{x}, \boldsymbol{k}, t)\,\mathrm{d}\boldsymbol{k} + \sum_{\alpha=\mathrm{S,P}} \int_{\mathbb{R}^2} \mathrm{Tr}\,\boldsymbol{W}_\alpha^+(\boldsymbol{x}, \boldsymbol{k}, t)\,\mathrm{d}\boldsymbol{k}, \tag{23}$$

while the power flow density vector is estimated by

$$\boldsymbol{\Pi}(\boldsymbol{x}, t) = c_{\mathrm{T}}(\boldsymbol{x}) \int_{\mathbb{R}^2} w_{\mathrm{T}}^+(\boldsymbol{x}, \boldsymbol{k}, t)\hat{\boldsymbol{k}}\,\mathrm{d}\boldsymbol{k} + \sum_{\alpha=\mathrm{S,P}} c_\alpha(\boldsymbol{x}) \int_{\mathbb{R}^2} \mathrm{Tr}\,\boldsymbol{W}_\alpha^+(\boldsymbol{x}, \boldsymbol{k}, t)\hat{\boldsymbol{k}}\,\mathrm{d}\boldsymbol{k}. \tag{24}$$

The rays $\tau \mapsto \boldsymbol{x}_\alpha(\tau)$ for each mode α are characterized by their tangent vector $\frac{\mathrm{d}\boldsymbol{x}_\alpha}{\mathrm{d}\tau} = c_\alpha \boldsymbol{t}$ with $\left|\frac{\mathrm{d}\boldsymbol{x}_\alpha}{\mathrm{d}\tau}\right| = 1$ and $\frac{\mathrm{d}\boldsymbol{t}}{\mathrm{d}\tau} = -|\boldsymbol{t}|\nabla_\Sigma c_\alpha$; after eliminating $\boldsymbol{t}$, a generalized Snell–Descartes law is obtained in the form

$$\frac{\mathrm{d}}{\mathrm{d}\tau}\left(\frac{1}{c_\alpha}\frac{\mathrm{d}\boldsymbol{x}_\alpha}{\mathrm{d}\tau}\right) = \nabla_\Sigma\left(\frac{1}{c_\alpha}\right). \tag{25}$$

In the case of an homogeneous material where c_α is independent of position $\boldsymbol{x}$, the above equation shows that the rays are the geodesic lines of the shell. Finally, the time τ_{12} needed by the energy density to travel the rays from a point $\boldsymbol{x}_1$ to a point $\boldsymbol{x}_2$ is given by $\tau_{12} = \int_{\boldsymbol{x}_1}^{\boldsymbol{x}_2} |\boldsymbol{t}|\,\mathrm{d}\boldsymbol{x}$.

4. The random case: radiative transfer

In this section we consider some randomness of the mechanical properties of the shell. We show how transport equations (20) and (21) are modified to account for scattering of the energy rays on random heterogeneities. This scattering mechanism is responsible for the randomization of the directions of rays which leads, after long propagation times, to the diffusion regime. Section 5 describes how it arises starting with the results presented now. As regards the terminology, transport equations modified to account for the scattering effects are called *radiative transfer equations* in the remaining of the paper.

4.1. General considerations

We follow again the basic developments of Papanicolaou and Ryzhik [23]. In their work these authors introduce a random perturbation $(\boldsymbol{V}(\boldsymbol{x}), \boldsymbol{x} \in \mathbb{R}^d)$ of the matrix $\boldsymbol{M}$ of the form:

$$\boldsymbol{M}(\boldsymbol{x}) = \underline{\boldsymbol{M}}(\boldsymbol{x})\left[\boldsymbol{I}_n + \sqrt{\epsilon}\boldsymbol{V}\left(\frac{\boldsymbol{x}}{\epsilon}\right)\right]. \tag{26}$$

The magnitude $\sqrt{\epsilon}$ of perturbations is chosen such that they have comparable effects to the unperturbed background medium characterized by the mean matrix of parameters $\underline{\boldsymbol{M}}(\boldsymbol{x})$. However they need not be too strong, for in this case localization rather than transport may occur. Their correlation lengths are chosen to be comparable to wavelengths ϵ so that complex multiple scattering phenomena may arise. The parameter fluctuations are modelized by a statistically homogeneous, second-order matrix-valued random process with mean zero whose covariance functions are denoted by $R_{jklm}(\boldsymbol{x})$ and such that

$$R_{jklm}(\boldsymbol{x} - \boldsymbol{y}) = \mathbb{E}\{V_{jk}(\boldsymbol{x})V_{lm}(\boldsymbol{y})\} = \int_{\mathbb{R}^d} \mathrm{e}^{-\mathrm{i}\boldsymbol{k}\cdot(\boldsymbol{x}-\boldsymbol{y})}\hat{R}_{jklm}(\boldsymbol{k})\,\mathrm{d}\boldsymbol{k} \tag{27}$$

where $\mathbb{E}\{\cdot\}$ stands for mathematical expectation. $\boldsymbol{k} \mapsto \hat{R}_{jklm}(\boldsymbol{k})$ are the spectral density functions of random matrix $(\boldsymbol{V}(\boldsymbol{x}), \boldsymbol{x} \in \mathbb{R}^d)$ for $j, k, l, m \in \{1, 2, \ldots, n\}$. They satisfy the usual properties of spectral densities of homogeneous stochastic processes, that is

$$\hat{R}_{jklm}(\boldsymbol{k}) = \overline{\hat{R}_{lmjk}(\boldsymbol{k})} = \hat{R}_{lmjk}(-\boldsymbol{k}). \tag{28}$$

Let $\hat{\boldsymbol{R}}$ be the fourth-order spectral density tensor corresponding to the above spectral density functions. Then it can be shown that the coherence matrices $\boldsymbol{W}_\alpha$, $1 \leqslant \alpha \leqslant M$, satisfy coupled radiative transfer equations given by

$$\begin{aligned}\partial_t \boldsymbol{W}_\alpha + \{\lambda_\alpha, \boldsymbol{W}_\alpha\} = \sum_{\beta=1}^{M} \int_{\mathbb{R}^d} \sigma_{\alpha\beta}(\boldsymbol{k}, \boldsymbol{k}')\boldsymbol{W}_\beta(\boldsymbol{k}')\delta(\lambda_\alpha(\boldsymbol{k}) - \lambda_\beta(\boldsymbol{k}'))\,\mathrm{d}\boldsymbol{k}' \\ + [\boldsymbol{Q}_\alpha, \boldsymbol{W}_\alpha] - [\boldsymbol{\Sigma}_\alpha, \boldsymbol{W}_\alpha]. \end{aligned} \tag{29}$$

The differential scattering operators $\sigma_{\alpha\beta}$ are defined by

$$\sigma_{\alpha\beta}(\boldsymbol{k}, \boldsymbol{k}')\boldsymbol{W}_\beta(\boldsymbol{k}') = 2\pi\lambda_\alpha^2(\boldsymbol{k}) \times \mathcal{S}_{\alpha\beta}(\boldsymbol{k}, \boldsymbol{k}')\boldsymbol{W}_\beta(\boldsymbol{k}') \tag{30}$$

where $\mathcal{S}_{\alpha\beta}(\boldsymbol{k}, \boldsymbol{k}')$ is a fourth-order tensor with elements

$$\mathcal{S}_{rstu}^{\alpha\beta}(\boldsymbol{k}, \boldsymbol{k}') = \boldsymbol{c}_{\alpha_r}(\boldsymbol{k}) \otimes \boldsymbol{b}_{\beta_t}(\boldsymbol{k}') : \hat{\boldsymbol{R}}(\boldsymbol{k} - \boldsymbol{k}') : \boldsymbol{c}_{\alpha_s}(\boldsymbol{k}) \otimes \boldsymbol{b}_{\beta_u}(\boldsymbol{k}'). \tag{31}$$

The total scattering operators $\boldsymbol{\Sigma}_\alpha$ are defined by

$$\begin{aligned}\boldsymbol{\Sigma}_\alpha(\boldsymbol{k}) = \frac{1}{2}\sum_{\beta=1}^{M} \Bigg\{ \int_{\mathbb{R}^d} \sigma_{\alpha\beta}(\boldsymbol{k}, \boldsymbol{k}')\boldsymbol{I}_{R_\beta}\delta(\lambda_\alpha(\boldsymbol{k}) - \lambda_\beta(\boldsymbol{k}'))\,\mathrm{d}\boldsymbol{k}' \\ - \mathrm{i}\int_{\mathbb{R}^d} \frac{1}{\lambda_\alpha(\boldsymbol{k}) - \lambda_\beta(\boldsymbol{k}')}\sigma_{\alpha\beta}(\boldsymbol{k}, \boldsymbol{k}')\boldsymbol{I}_{R_\beta}\,\mathrm{d}\boldsymbol{k}' \Bigg\}. \end{aligned} \tag{32}$$

$\boldsymbol{b}_{\alpha_r}$ are the right eigenvectors of the mean dispersion matrix and $\boldsymbol{c}_{\alpha_r}$ are its left eigenvectors, given by $\boldsymbol{c}_{\alpha_r} = \underline{\boldsymbol{M}}\boldsymbol{b}_{\alpha_r}$, for $1 \leqslant \alpha \leqslant M$ and $1 \leqslant r \leqslant R_\alpha$. The differential scattering operator $\sigma_{\alpha\beta}(\boldsymbol{x}, \boldsymbol{k}, \boldsymbol{k}')$ gives the rate at which the energy mode β with wave vector $\boldsymbol{k}'$ is converted to the energy mode α with wave vector $\boldsymbol{k}$, at position $\boldsymbol{x}$; the total scattering operator $\boldsymbol{\Sigma}_\alpha(\boldsymbol{x}, \boldsymbol{k})$ gives the rate at which the energy mode α with wave vector $\boldsymbol{k}$ at position $\boldsymbol{x}$ is converted to all other modes or wave numbers (thus it can account for intrinsic damping).

4.2. *Radiative transfer equations for a cylindrical shell*

Let us consider a random perturbation matrix $(\boldsymbol{V}(\boldsymbol{x}), \boldsymbol{x} \in \mathbb{R}^2)$ having the form

$$\boldsymbol{V}(\boldsymbol{x}) = \begin{bmatrix} \tilde{\varrho}(\boldsymbol{x})\boldsymbol{I}_5 & 0 & 0 & 0 \\ 0 & \tilde{\psi}(\boldsymbol{x})\boldsymbol{I}_3 & 0 & 0 \\ 0 & 0 & \tilde{\chi}(\boldsymbol{x})\boldsymbol{I}_2 & 0 \\ 0 & 0 & 0 & \tilde{\psi}(\boldsymbol{x})\boldsymbol{I}_3 \end{bmatrix}$$

where $(\tilde{\varrho}(\boldsymbol{x}), \boldsymbol{x} \in \mathbb{R}^2)$, $(\tilde{\chi}(\boldsymbol{x}), \boldsymbol{x} \in \mathbb{R}^2)$ and $(\tilde{\psi}(\boldsymbol{x}), \boldsymbol{x} \in \mathbb{R}^2)$ are three real-valued, statistically homogeneous and centred stochastic processes whose spectral measures are characterized by their power spectral density functions $\boldsymbol{k} \mapsto \hat{R}_\varrho(\boldsymbol{k})$, $\boldsymbol{k} \mapsto \hat{R}_\chi(\boldsymbol{k})$ and $\boldsymbol{k} \mapsto \hat{R}_\psi(\boldsymbol{k})$, respectively, and cross-spectral density functions $\boldsymbol{k} \mapsto \hat{R}_{\varrho\chi}(\boldsymbol{k})$, $\boldsymbol{k} \mapsto \hat{R}_{\varrho\psi}(\boldsymbol{k})$ and $\boldsymbol{k} \mapsto \hat{R}_{\chi\psi}(\boldsymbol{k})$. These processes represent dimensionless random perturbations of parameters ϱ, $(\kappa G)^{-1}$ and E^{-1}, respectively, such that they write

$$\begin{aligned} \varrho(\boldsymbol{x}) &= \underline{\varrho}(\boldsymbol{x})\left[1 + \sqrt{\epsilon}\tilde{\varrho}\left(\tfrac{\boldsymbol{x}}{\epsilon}\right)\right] \\ E^{-1}(\boldsymbol{x}) &= \underline{E^{-1}}(\boldsymbol{x})\left[1 + \sqrt{\epsilon}\tilde{\psi}\left(\tfrac{\boldsymbol{x}}{\epsilon}\right)\right] \\ (\kappa G)^{-1}(\boldsymbol{x}) &= \underline{(\kappa G)^{-1}}(\boldsymbol{x})\left[1 + \sqrt{\epsilon}\tilde{\chi}\left(\tfrac{\boldsymbol{x}}{\epsilon}\right)\right], \end{aligned}$$

where $\underline{\varrho}$, $\underline{E^{-1}}$ and $\underline{(\kappa G)^{-1}}$ are their mean values. We assume that shell thickness h and Poisson's coefficient ν remain deterministic, but the latter hypothesis is not restrictive. Then the differential scattering operators are simple multiplicative operators characterized by scalars, of which expressions are given as [33]: for the propagative shear mode T

$$\sigma_{\mathrm{TT}}(\boldsymbol{k},\boldsymbol{k}') = \frac{\pi}{2}c_{\mathrm{T}}^2|\boldsymbol{k}|^2[\hat{R}_{\varrho}(\boldsymbol{k}-\boldsymbol{k}') + (\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')^2\hat{R}_{\chi}(\boldsymbol{k}-\boldsymbol{k}') + 2(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')\operatorname{Re}\hat{R}_{\varrho\chi}(\boldsymbol{k}-\boldsymbol{k}')], \tag{33}$$

and for the propagative bending modes P and S

$$\begin{aligned}\sigma_{\mathrm{PP}}(\boldsymbol{k},\boldsymbol{k}') &= \frac{\pi}{2}c_{\mathrm{P}}^2|\boldsymbol{k}|^2[(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')^2\hat{R}_{\varrho}(\boldsymbol{k}-\boldsymbol{k}') + [(1-\nu)(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')^2+\nu]^2\hat{R}_{\psi}(\boldsymbol{k}-\boldsymbol{k}')\\ &\quad + 2(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')[(1-\nu)(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')^2+\nu]\operatorname{Re}\hat{R}_{\varrho\psi}(\boldsymbol{k}-\boldsymbol{k}')],\end{aligned} \tag{34}$$

$$\begin{aligned}\sigma_{\mathrm{SS}}(\boldsymbol{k},\boldsymbol{k}') &= \frac{\pi}{2}c_{\mathrm{S}}^2|\boldsymbol{k}|^2[(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')^2\hat{R}_{\varrho}(\boldsymbol{k}-\boldsymbol{k}') + [2(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')^2-1]^2\hat{R}_{\psi}(\boldsymbol{k}-\boldsymbol{k}')\\ &\quad + 2(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')[2(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')^2-1]\operatorname{Re}\hat{R}_{\varrho\psi}(\boldsymbol{k}-\boldsymbol{k}')],\end{aligned} \tag{35}$$

$$\begin{aligned}\sigma_{\mathrm{PS}}(\boldsymbol{k},\boldsymbol{k}') &= \frac{\pi}{2}c_{\mathrm{S}}^2(1-(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')^2)[|\boldsymbol{k}'|^2\hat{R}_{\varrho}(\boldsymbol{k}-\boldsymbol{k}') + 4|\boldsymbol{k}|^2(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')^2\hat{R}_{\psi}(\boldsymbol{k}-\boldsymbol{k}')\\ &\quad + 4|\boldsymbol{k}||\boldsymbol{k}'|(\hat{\boldsymbol{k}}\cdot\hat{\boldsymbol{k}}')\operatorname{Re}\hat{R}_{\varrho\psi}(\boldsymbol{k}-\boldsymbol{k}')],\end{aligned} \tag{36}$$

with $\sigma_{\mathrm{SP}}(\boldsymbol{x},\boldsymbol{k},\boldsymbol{k}') = \sigma_{\mathrm{PS}}(\boldsymbol{x},\boldsymbol{k}',\boldsymbol{k})$. Thus forward (+) and backward (−) going waves are uncoupled, as well as the pure shear mode with respect to the bending modes. The total scattering cross-sections become scalars given by

$$\Sigma_{\alpha}(\boldsymbol{x},\boldsymbol{k}) = \sum_{\beta}\Sigma_{\alpha\beta}(\boldsymbol{x},\boldsymbol{k}), \qquad \Sigma_{\alpha\beta}(\boldsymbol{x},\boldsymbol{k}) = \int_{\mathbb{R}^2}\sigma_{\alpha\beta}(\boldsymbol{x},\boldsymbol{k},\boldsymbol{k}')\delta(c_{\alpha}|\boldsymbol{k}|-c_{\beta}|\boldsymbol{k}'|)\,\mathrm{d}\boldsymbol{k}'. \tag{37}$$

The multigroup radiative transfer equations satisfied by coherence matrices $\boldsymbol{W}_{\mathrm{P}}^{\pm}$ and $\boldsymbol{W}_{\mathrm{S}}^{\pm}$ are

$$\begin{aligned}\partial_t\boldsymbol{W}_{\mathrm{P}}^{\pm} + \left\{\lambda_{\mathrm{P}}^{\pm},\boldsymbol{W}_{\mathrm{P}}^{\pm}\right\} &= \int_{\mathbb{R}^2}\sigma_{\mathrm{PP}}(\boldsymbol{x},\boldsymbol{k},\boldsymbol{k}')\boldsymbol{W}_{\mathrm{P}}^{\pm}(\boldsymbol{x},\boldsymbol{k}',t)\delta(c_{\mathrm{P}}|\boldsymbol{k}|-c_{\mathrm{P}}|\boldsymbol{k}'|)\,\mathrm{d}\boldsymbol{k}'\\ &\quad + \int_{\mathbb{R}^2}\sigma_{\mathrm{PS}}(\boldsymbol{x},\boldsymbol{k},\boldsymbol{k}')\boldsymbol{W}_{\mathrm{S}}^{\pm}(\boldsymbol{x},\boldsymbol{k}',t)\delta(c_{\mathrm{P}}|\boldsymbol{k}|-c_{\mathrm{S}}|\boldsymbol{k}'|)\,\mathrm{d}\boldsymbol{k}'\\ &\quad - \Sigma_{\mathrm{P}}(\boldsymbol{x},\boldsymbol{k})\boldsymbol{W}_{\mathrm{P}}^{\pm}(\boldsymbol{x},\boldsymbol{k},t)\end{aligned} \tag{38}$$

and

$$\begin{aligned}\partial_t\boldsymbol{W}_{\mathrm{S}}^{\pm} + \left\{\lambda_{\mathrm{S}}^{\pm},\boldsymbol{W}_{\mathrm{S}}^{\pm}\right\} &= \int_{\mathbb{R}^2}\sigma_{\mathrm{SS}}(\boldsymbol{x},\boldsymbol{k},\boldsymbol{k}')\boldsymbol{W}_{\mathrm{S}}^{\pm}(\boldsymbol{x},\boldsymbol{k}',t)\delta(c_{\mathrm{S}}|\boldsymbol{k}|-c_{\mathrm{S}}|\boldsymbol{k}'|)\,\mathrm{d}\boldsymbol{k}'\\ &\quad + \int_{\mathbb{R}^2}\sigma_{\mathrm{SP}}(\boldsymbol{x},\boldsymbol{k},\boldsymbol{k}')\boldsymbol{W}_{\mathrm{P}}^{\pm}(\boldsymbol{x},\boldsymbol{k}',t)\delta(c_{\mathrm{S}}|\boldsymbol{k}|-c_{\mathrm{P}}|\boldsymbol{k}'|)\,\mathrm{d}\boldsymbol{k}'\\ &\quad - \Sigma_{\mathrm{S}}(\boldsymbol{x},\boldsymbol{k})\boldsymbol{W}_{\mathrm{S}}^{\pm}(\boldsymbol{x},\boldsymbol{k},t),\end{aligned} \tag{39}$$

while the scalar radiative transfer equation satisfied by the specific intensity $w_{\mathrm{T}}^{\pm}$ is

$$\begin{aligned}\partial_t w_{\mathrm{T}}^{\pm} + \left\{\lambda_{\mathrm{T}}^{\pm},w_{\mathrm{T}}^{\pm}\right\} &= \int_{\mathbb{R}^2}\sigma_{\mathrm{TT}}(\boldsymbol{x},\boldsymbol{k},\boldsymbol{k}')w_{\mathrm{T}}^{\pm}(\boldsymbol{x},\boldsymbol{k}',t)\delta(c_{\mathrm{T}}|\boldsymbol{k}|-c_{\mathrm{T}}|\boldsymbol{k}'|)\,\mathrm{d}\boldsymbol{k}'\\ &\quad - \Sigma_{\mathrm{T}}(\boldsymbol{x},\boldsymbol{k})w_{\mathrm{T}}^{\pm}(\boldsymbol{x},\boldsymbol{k},t).\end{aligned} \tag{40}$$

4.3. Numerical examples: scattering cross-sections

If we assume that random matrix $(\boldsymbol{V}(\boldsymbol{x}),\,\boldsymbol{x}\in\mathbb{R}^2)$ is statistically isotropic, the spectral density functions $\boldsymbol{k}\mapsto\hat{R}_{jk}(\boldsymbol{k})$, $j,k\in\{\varrho,\chi,\psi\}$, are all real and depend on $|\boldsymbol{k}|$ rather than $\boldsymbol{k}$. Then

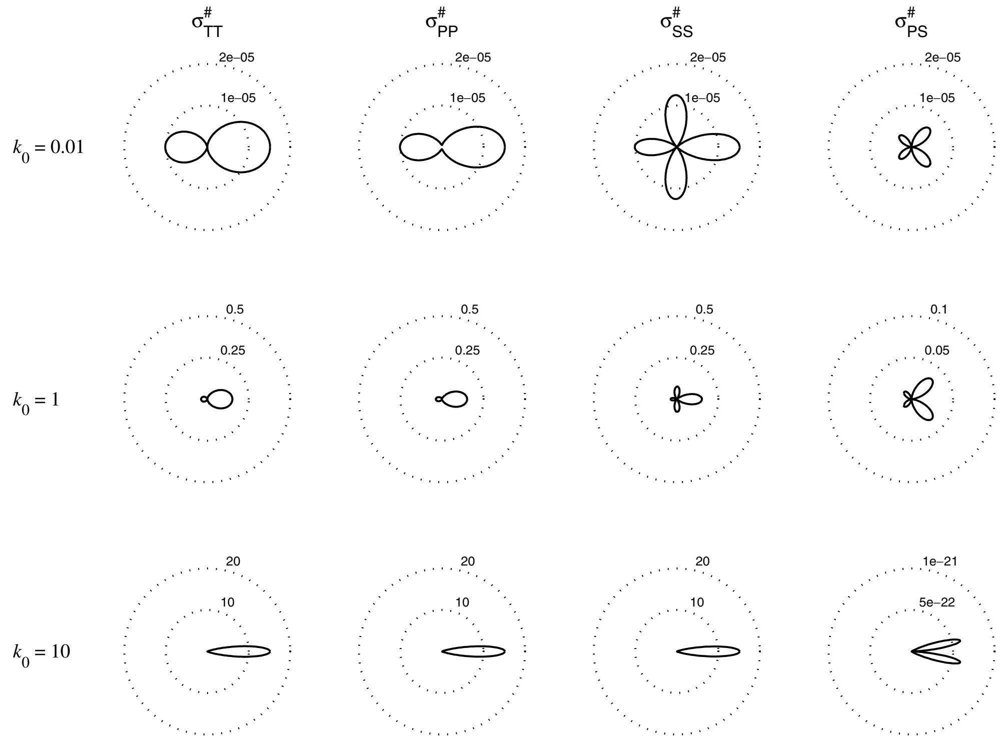

Figure 2. Normalized differential scattering cross-sections for a thick shell with Gaussian correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$ and $\xi_\chi = \xi_\psi = 1$.

scattering cross-sections are rotationally invariant, which means that they depend on $|\boldsymbol{k}|$ and $\hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}}' = \cos\Theta$ solely. The normalized differential scattering cross-sections $\sigma^{\#}_{\alpha\beta}$ defined by

$$\sigma^{\#}_{\alpha\beta}(|\boldsymbol{k}|, \Theta) := \frac{1}{c_\alpha^2}\sigma_{\alpha\beta}(\boldsymbol{k}, \boldsymbol{k}'), \qquad \alpha, \beta \in \{\mathrm{T}, \mathrm{P}, \mathrm{S}\},$$

where the $\sigma_{\alpha\beta}$ are given by equations (33) through (36), are plotted in polar coordinates as functions of Θ on figure 2 and figure 4 for Gaussian correlations of the perturbations, and on figure 3 and figure 5 for exponential correlations. For Gaussian covariance functions

$$R_{jk}(\boldsymbol{x} - \boldsymbol{y}) = \xi_j \xi_k \mathrm{e}^{-\gamma^2|\boldsymbol{x}-\boldsymbol{y}|^2}$$

we have

$$\hat{R}_{jk}(|\boldsymbol{k}|) = \frac{\xi_j \xi_k}{4\pi\gamma^2} \exp\left(-\frac{|\boldsymbol{k}|^2}{4\gamma^2}\right),$$

while for exponential covariance functions

$$R_{jk}(\boldsymbol{x} - \boldsymbol{y}) = \xi_j \xi_k \mathrm{e}^{-\gamma|\boldsymbol{x}-\boldsymbol{y}|}$$

we have

$$\hat{R}_{jk}(|\boldsymbol{k}|) = \frac{\xi_j \xi_k}{2\pi\gamma^2} \left(1 + \frac{|\boldsymbol{k}|^2}{\gamma^2}\right)^{-\frac{3}{2}}.$$

γ^{-1} is a correlation lengthscale of the perturbations and the ξ_j, $j \in \{\varrho, \chi, \psi\}$, stand for the fluctuation amplitudes of parameters. Figures 2 and 3 correspond to $\xi_\varrho = 0 \cdot 1$ and

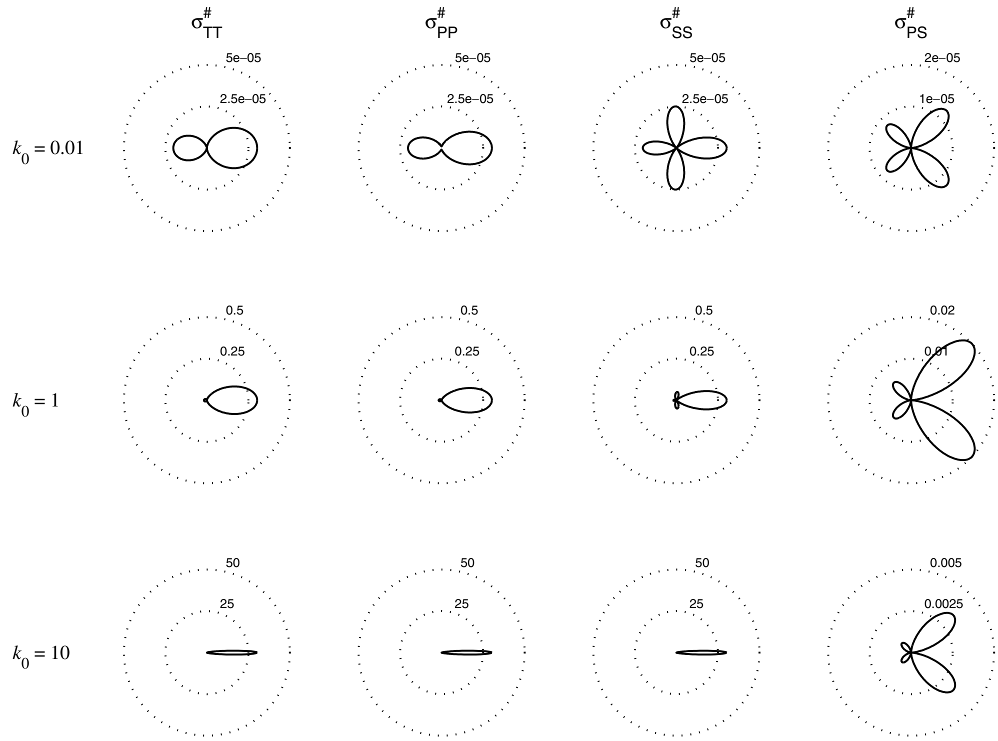

Figure 3. Normalized differential scattering cross-sections for a thick shell with exponential correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$ and $\xi_\chi = \xi_\psi = 1$.

$\xi_\chi = \xi_\psi = 1$, while figures 4 and 5 correspond to $\xi_\varrho = 1$ and $\xi_\chi = \xi_\psi = 0 \cdot 1$. The dimensionless wave number is $k_0 = \gamma^{-1}|\boldsymbol{k}|$. It is seen that forward scattering dominates at almost all frequencies, and becomes even more pronounced as frequency increases. We also outline the overall strong anisotropy—even at low frequencies, except for σ_{TT} and a strongly fluctuating mass—of the differential scattering cross-sections, despite the assumption of statistical isotropy. Changes in the fluctuation scales affect them only at low frequencies. Observe that P to S or S to P mode conversion is forbidden in the exact forward and backward directions, while S to S scattering may occur in the transverse direction at low frequencies and for strong fluctuations of the stiffness (see figures 2 and 3).

Figure 6 displays the normalized total scattering cross-sections $\Sigma^{\#}_{\alpha\beta} = \Sigma_{\alpha\beta}/\gamma c_\alpha$, $\alpha, \beta \in \{\mathrm{T, P, S}\}$ and $\Sigma_{\alpha\beta}$ given by equation (37), as functions of k_0 for Gaussian correlations, and figure 7 displays them for exponential correlations; here we have chosen $\xi_\varrho = 0 \cdot 1$ and $\xi_\chi = \xi_\psi = 1$ since the mass is known to fluctuate much less than the stiffness. Note that they depend on $|\boldsymbol{k}|$ solely and that $\Sigma_{\mathrm{SP}}(|\boldsymbol{k}|) = \left(\frac{c_{\mathrm{S}}}{c_{\mathrm{P}}}\right)^2 \Sigma_{\mathrm{PS}}\left(\frac{c_{\mathrm{S}}}{c_{\mathrm{P}}}|\boldsymbol{k}|\right)$. Finally for all plots Poisson's coefficient has been chosen as $\nu = 0 \cdot 2$.

4.4. Numerical examples: resolution of radiative transfer equations by the Monte Carlo method

Numerical solutions of linear radiative transfer equations can be constructed by the direct Monte Carlo method described in the book of Lapeyre *et al* [31], for instance. It consists in

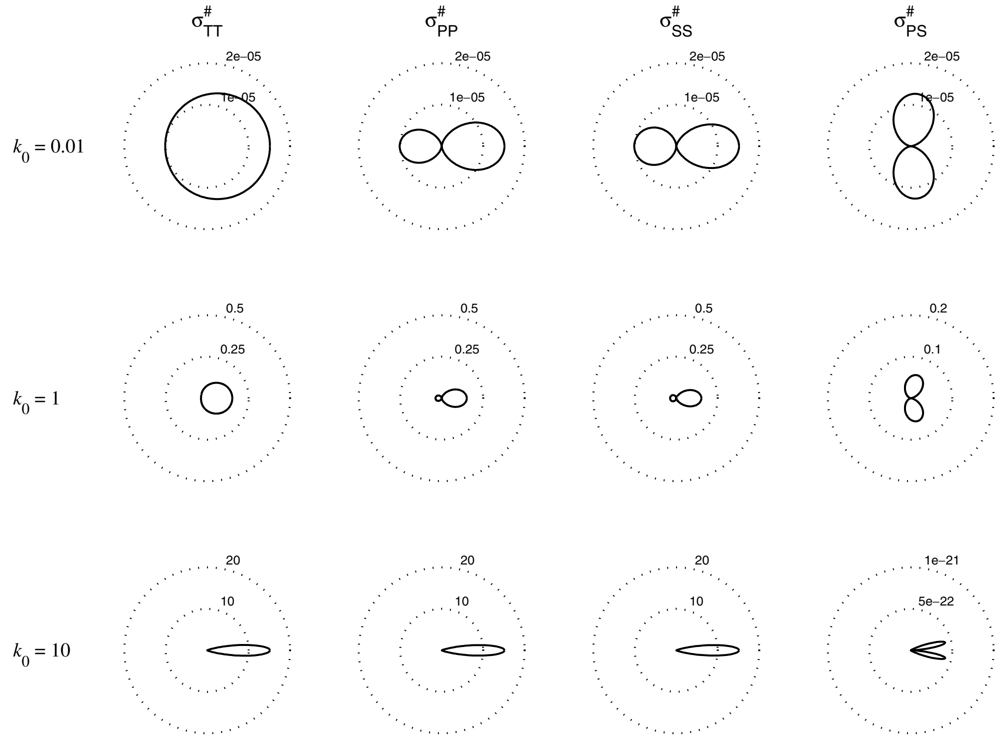

Figure 4. Normalized differential scattering cross-sections for a thick shell with Gaussian correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 1$ and $\xi_\chi = \xi_\psi = 0{\cdot}1$.

simulating particle trajectories of a particular class of transport processes and then averaging over these paths. We consider here the case of an homogeneous constant-velocity background medium with statistically isotropic random heterogeneities on the scale of the wavelength. Given the structure of the multigroup radiative transfer equations (38) and (39) for bending, and the scalar one (40) for pure shear, and since we focus on the energy density (23), this system of equations is equivalent to

$$\partial_t \boldsymbol{w}(\boldsymbol{x}, \boldsymbol{k}, t) + \boldsymbol{c}\hat{\boldsymbol{k}} \cdot \nabla_\Sigma \boldsymbol{w}(\boldsymbol{x}, \boldsymbol{k}, t) = \int_{S^1} \tilde{\boldsymbol{\sigma}}(|\boldsymbol{k}|, \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}}')\boldsymbol{w}(\boldsymbol{x}, \boldsymbol{k}', t)\, \mathrm{d}\hat{\boldsymbol{k}}' - \boldsymbol{\Sigma}(\boldsymbol{k})\boldsymbol{w}(\boldsymbol{x}, \boldsymbol{k}, t) \quad (41)$$

where $\boldsymbol{c} = \mathrm{diag}(c_\mathrm{T}, c_\mathrm{P}, c_\mathrm{S})$ is the velocity matrix, $\boldsymbol{\Sigma} = \mathrm{diag}(\Sigma_\mathrm{T}, \Sigma_\mathrm{P}, \Sigma_\mathrm{S})$ is the total scattering matrix and $\tilde{\boldsymbol{\sigma}}$ is the kernel of the collision operator

$$\tilde{\boldsymbol{\sigma}} = \begin{bmatrix} \tilde{\sigma}_{\mathrm{TT}} & 0 & 0 \\ 0 & \tilde{\sigma}_{\mathrm{PP}} & \tilde{\sigma}_{\mathrm{PS}} \\ 0 & \tilde{\sigma}_{\mathrm{SP}} & \tilde{\sigma}_{\mathrm{SS}} \end{bmatrix}, \qquad \sigma_{\alpha\beta}(\boldsymbol{k}, \boldsymbol{k}') := \frac{c_\beta}{|\boldsymbol{k}'|}\tilde{\sigma}_{\alpha\beta}(|\boldsymbol{k}|, \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}}'). \quad (42)$$

$\boldsymbol{w} = (w_\mathrm{T}, w_\mathrm{P}, w_\mathrm{S})$ is the vector of the angularly resolved specific intensities for shear—mode labelled T—and bending—modes labelled P and S—motions with $w_\mathrm{P} := \mathrm{Tr}\, \boldsymbol{W}_\mathrm{P}^+$ and $w_\mathrm{S} := \mathrm{Tr}\, \boldsymbol{W}_\mathrm{S}^+$. At last S^1 stands for the unit circle of $\mathbb{R}^2$. The test structure considered here is a circular cylindrical shell of radius R and length L (see figure 8). Boundary conditions at $x_1 = \pm\frac{L}{2}$ are taken into account in our simulations with the assumption that the paths are

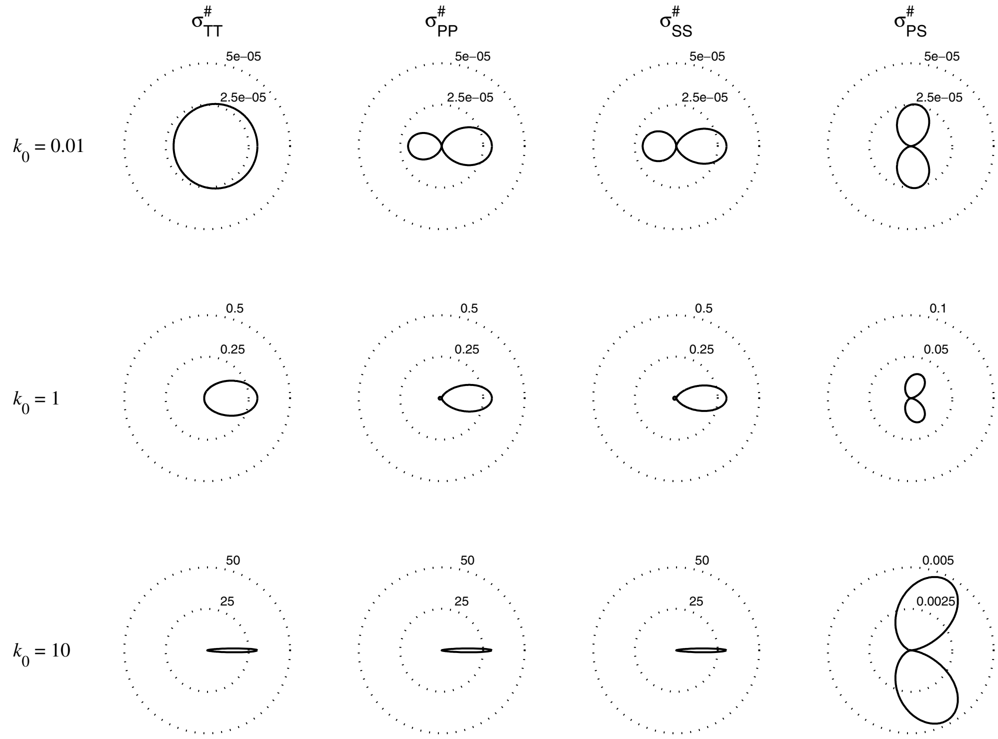

Figure 5. Normalized differential scattering cross-sections for a thick shell with exponential correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 1$ and $\xi_\chi = \xi_\psi = 0{\cdot}1$.

reflected according to Snell–Descartes' law (lossless reflections). The latter leads to

$$\boldsymbol{w}(\boldsymbol{x}, \boldsymbol{k}, t) = \boldsymbol{w}(\boldsymbol{x}, \boldsymbol{k} - 2(\boldsymbol{k} \cdot \boldsymbol{e}_1)\boldsymbol{e}_1, t)$$

when $\boldsymbol{x}$ is a point on a boundary line of the shell at $x_1 = \pm\frac{L}{2}$. Here we also neglect the possible mode conversions at the boundaries, not because they are unlikely to occur but because we lack a theoretical model of such a scattering mechanism for energetic quantities.

Figure 9 displays our results for bending energetic waves with the Gaussian correlation of the perturbations, $\xi_\varrho = 0 \cdot 1$ and $\xi_\chi = \xi_\psi = 1$ for the parameters fluctuation amplitudes, $\nu = 0 \cdot 2$ for Poisson's coefficient, $c_\mathrm{S} = 1\mathrm{m\,s}^{-1}$ and $k_0 = 1$. Figure 10 is the same for $k_0 = 10$. A unit point source is applied at time $t = 0$ near one extremity of the shaft and it equally loads both bending modes P and S. The radius R and length L of the shell are one half and six times, respectively, the maximum transport mean free paths $\max(\ell^*_\mathrm{P}, \ell^*_\mathrm{S})$ defined in the next section. Time steps are multiples of the diffusion time $t^*_\mathrm{PS} = \max(\ell^*_\mathrm{P}/c_\mathrm{P}, \ell^*_\mathrm{S}/c_\mathrm{S})$.

5. Diffusive regime for cylindrical shells

Transported waves exhibit a diffusive behaviour inasmuch as their direction of propagation is randomized after travelling several scattering mean free paths, e.g., the mean distances they travel before their direction of propagation is significantly altered. They also become approximately depolarized in this regime. This situation corresponds to the conditions in which SEA is applied for *bounded* structures. In this case, the scattering mean free paths are

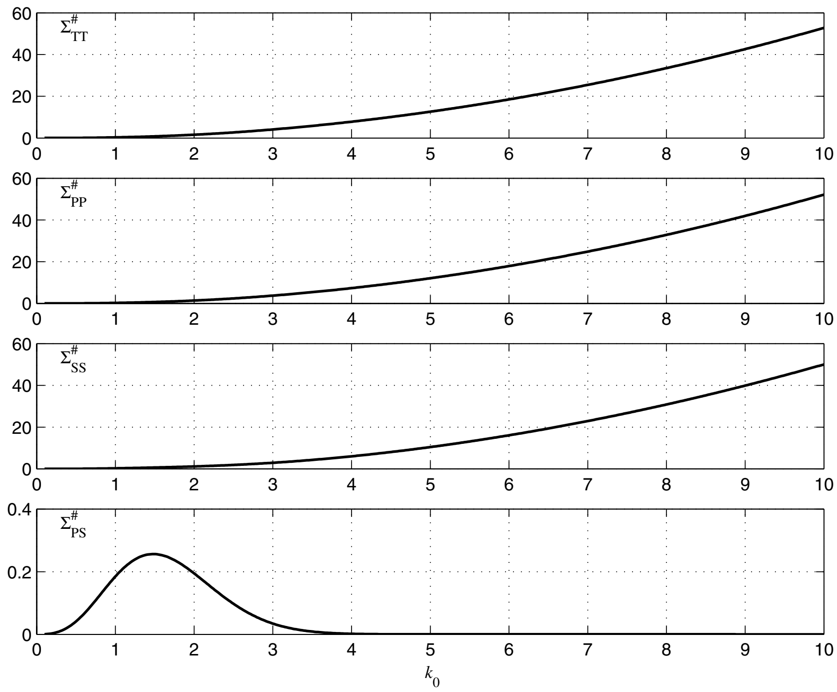

Figure 6. Normalized total scattering cross-sections for a thick shell with Gaussian correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$ and $\xi_\chi = \xi_\psi = 1$.

presumably small with respect to the characteristic dimensions of the structure. Proportionality relations between the energy densities of particular propagative modes and their associated velocities, similarly to the ones given between vibrational energies and modal densities in SEA, can be obtained in the diffusion limit. It should be noted, however, that in infinitely extended homogeneous media, scattering mean free paths are infinite and then a diffusive behaviour can theoretically never be reached. This is in contradiction with the usual assumptions used in the vibrational conductivity approach as it stands to date. The following results establish the correct setting for the validity of the heat conduction analogy in slender structures such as shells.

The scattering mean free path is defined for a particular propagating mode α by

$$\ell_\alpha(\boldsymbol{x}, \boldsymbol{k}) = \frac{|\boldsymbol{\nabla}_k \lambda_\alpha|}{\Sigma_\alpha(\boldsymbol{x}, \boldsymbol{k})}. \tag{43}$$

Derivations of the radiative transfer equations have been carried out so far for high frequencies such that $\Lambda \ll c_\alpha t$ and they remain applicable as long as $\Lambda \ll \ell_\alpha$, where Λ is a typical wavelength. The diffusion limit considers that propagation distances are furthermore much longer than the scattering mean free path: $c_\alpha t \gg \ell_\alpha$. It is characterized by a transport mean free path, denoted by ℓ^*_α for the mode α, which is the distance waves must travel until their direction is completely randomized. Along the same lines as in [23] for statistically isotropic random fluctuations of an homogeneous background medium, we obtain from equation (41) the following diffusive regime for our model of shells: (i) specific intensity w_T for pure shear

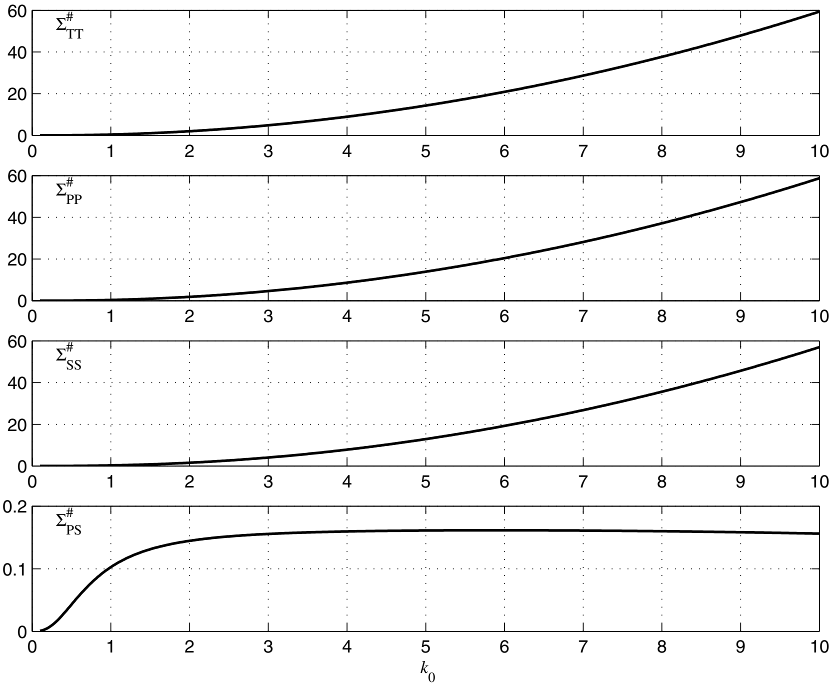

Figure 7. Normalized total scattering cross-sections for a thick shell with exponential correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$ and $\xi_\chi = \xi_\psi = 1$.

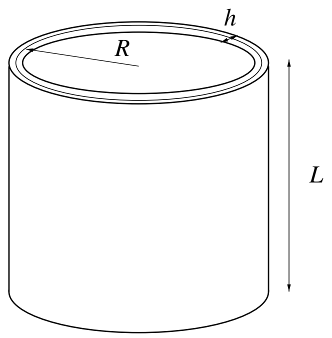

Figure 8. Circular cylindrical shell with radius R, length L and thickness h.

motion is independent of $\hat{\boldsymbol{k}}$ and satisfies the diffusion equation

$$\partial_t w_{\mathrm{T}}(\boldsymbol{x}, |\boldsymbol{k}|, t) = \boldsymbol{\nabla}_\Sigma \cdot [D_{\mathrm{T}}(|\boldsymbol{k}|)\boldsymbol{\nabla}_\Sigma w_{\mathrm{T}}(\boldsymbol{x}, |\boldsymbol{k}|, t)] \tag{44}$$

where $D_{\mathrm{T}} = \frac{1}{2} c_{\mathrm{T}} \ell^*_{\mathrm{T}}$ is the diffusion constant and ℓ^*_{T} is the shear transport mean free path given by

$$\ell^*(|\boldsymbol{k}|) = c_{\mathrm{T}} \left[\int_{S^1} \tilde{\sigma}_{\mathrm{TT}}(|\boldsymbol{k}|, \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}}')(1 - \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}}')\, \mathrm{d}\hat{\boldsymbol{k}}' \right]^{-1}. \tag{45}$$

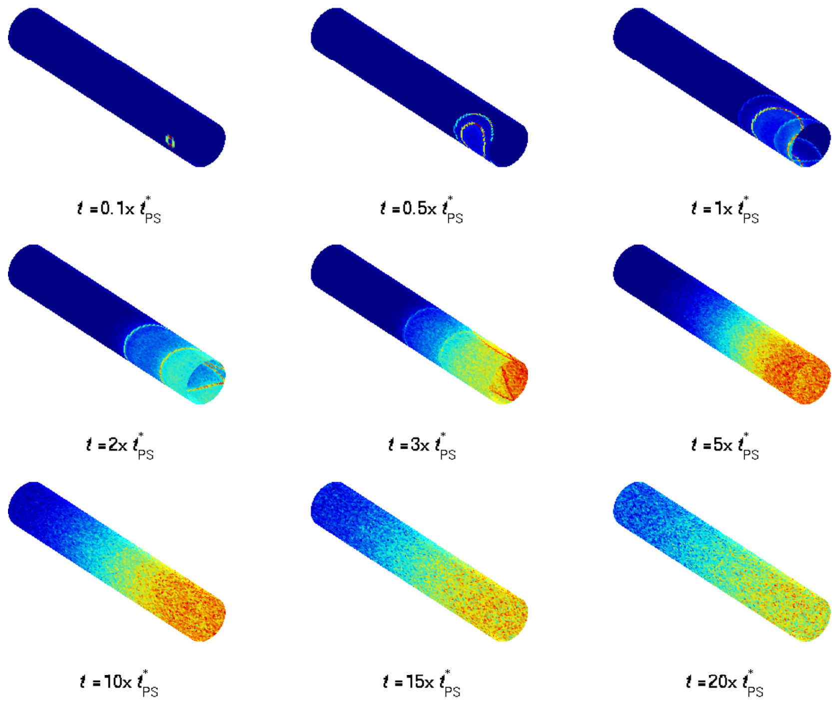

Figure 9. Bending energy density propagation with time in a circular cylindrical shell for $k_0 = 1$; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$, $\xi_\chi = \xi_\psi = 1$ and $c_S = 1\ \mathrm{m\ s^{-1}}$.

(ii) Specific intensities w_P and w_S describing bending motion are independent of $\hat{\boldsymbol{k}}$ as well and linked by an equipartition rule $w_P(\boldsymbol{x}, |\boldsymbol{k}|, t) = w_S\big(\boldsymbol{x}, \frac{c_P}{c_S}|\boldsymbol{k}|, t\big) := \Psi(\boldsymbol{x}, |\boldsymbol{k}|, t)$ such that

$$\partial_t \Psi(\boldsymbol{x}, |\boldsymbol{k}|, t) = \nabla_\Sigma \cdot [D_{PS}(|\boldsymbol{k}|)\nabla_\Sigma \Psi(\boldsymbol{x}, |\boldsymbol{k}|, t)]\,. \tag{46}$$

The diffusion coefficient is now

$$D_{PS}(|\boldsymbol{k}|) = \frac{c_P c_S}{2\big(c_P^2 + c_S^2\big)}\left(c_S \ell_P^*(|\boldsymbol{k}|) + c_P \ell_S^*\left(\frac{c_P}{c_S}|\boldsymbol{k}|\right)\right) \tag{47}$$

where the transport mean free paths ℓ_P^* and ℓ_S^* are the solutions of

$$\begin{bmatrix} \Sigma_P(|\boldsymbol{k}|) - \Sigma_{PP}^*(|\boldsymbol{k}|) & -\Sigma_{PS}^*(|\boldsymbol{k}|) \\ -\Sigma_{SP}^*\left(\frac{c_P}{c_S}|\boldsymbol{k}|\right) & \Sigma_S\left(\frac{c_P}{c_S}|\boldsymbol{k}|\right) - \Sigma_{SS}^*\left(\frac{c_P}{c_S}|\boldsymbol{k}|\right) \end{bmatrix} \begin{pmatrix} \ell_P^*(|\boldsymbol{k}|) \\ \ell_S^*\left(\frac{c_P}{c_S}|\boldsymbol{k}|\right) \end{pmatrix} = \begin{pmatrix} c_P \\ c_S \end{pmatrix}$$

with

$$\Sigma_{\alpha\beta}^*(|\boldsymbol{k}|) = \int_{S^1} \tilde{\sigma}_{\alpha\beta}(|\boldsymbol{k}|, \hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}}')\hat{\boldsymbol{k}} \cdot \hat{\boldsymbol{k}}'\, \mathrm{d}\hat{\boldsymbol{k}}', \qquad \alpha, \beta \in \{\mathrm{P}, \mathrm{S}\}.$$

Analytical expressions of the diffusion parameters (mean free paths, diffusion coefficients) are useful because these quantities can be estimated from experimental data, while correlation lengths and standard deviations of parameter fluctuations are usually not observable. They are now plotted in figure 11 through figure 14 for either Gaussian or exponential correlations. Figures 11 and 13 display the normalized scattering mean free paths $\ell_\alpha^\# = \gamma \ell_\alpha$ (dashed lines) and the normalized transport mean free paths $\ell_\alpha^{*\#} = \gamma \ell_\alpha^*$ (solid lines), $\alpha \in \{\mathrm{T}, \mathrm{P}, \mathrm{S}\}$,

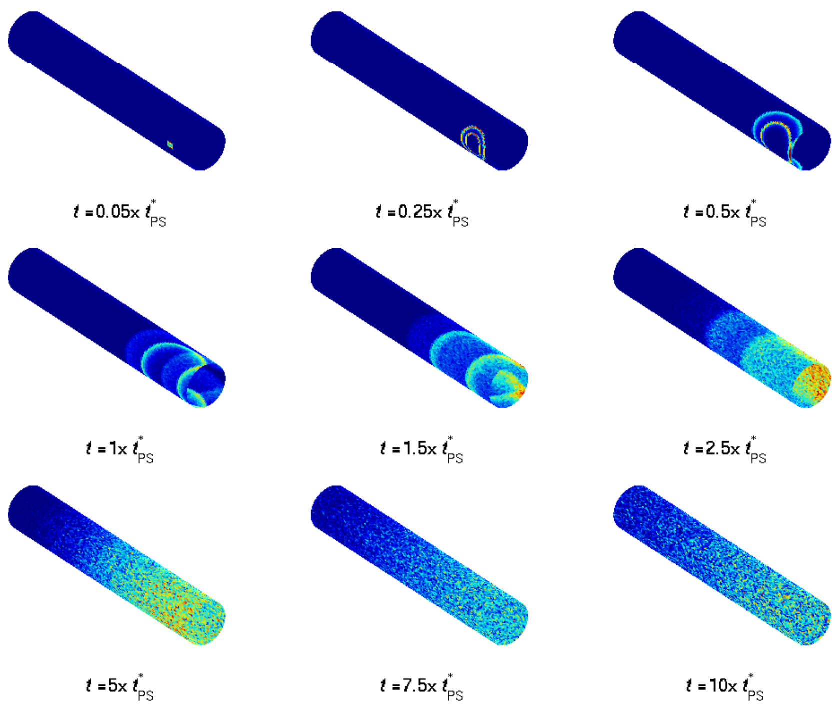

Figure 10. Bending energy density propagation with time in a circular cylindrical shell for $k_0 = 10$; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$, $\xi_\chi = \xi_\psi = 1$ and $c_S = 1$ m s^{-1}.

as functions of the dimensionless wave number k_0. The latter get larger than the former as frequency increases, as expected from the increased anisotropy of scattering—in the forward direction—seen in figure 2 through figure 5 as frequency increases. Also observe that transport mean free paths have nonzero, frequency-independent values over a broad frequency range: $\ell_\alpha^* \simeq 2\gamma^{-1}$ for $m = \mathrm{T}$, P and S, $k_0 = 10$ and Gaussian correlations, while $\ell_\alpha^* \simeq 1 \cdot 5\gamma^{-1}$ for exponential correlations. This is the so-called Rayleigh–Gans regime. The agreement of these results both qualitatively and quantitatively with the experimental data given by Schriemer *et al* [34] is striking, although they were obtained for a completely different medium (glass beads in water). Figures 12 and 14 display the normalized diffusion coefficient $D_{\mathrm{PS}}^\# = \gamma D_{\mathrm{PS}}/c_S$ as a function of the dimensionless wave number k_0. For all plots Poisson's coefficient is again $\nu = 0 \cdot 2$ while $\xi_\varrho = 0 \cdot 1$ and $\xi_\chi = \xi_\psi = 1$. The diffusion characteristics are only slightly affected by these fluctuation scales since forward scattering always dominates at high frequencies.

The equipartition rule of energies at late times is illustrated by figure 15. It displays the ratio of the bending energy $\mathcal{E}_{\mathrm{P}}(|\boldsymbol{k}|, t) = \int_\Sigma \int_{S^1} w_{\mathrm{P}}(\boldsymbol{x}, \boldsymbol{k}, t)\, \mathrm{d}\hat{\boldsymbol{k}}\, \mathrm{d}\boldsymbol{x}$ over the bending energy $\mathcal{E}_{\mathrm{S}}(|\boldsymbol{k}|, t) = \int_\Sigma \int_{S^1} w_{\mathrm{S}}(\boldsymbol{x}, \boldsymbol{k}, t)\, \mathrm{d}\hat{\boldsymbol{k}}\, \mathrm{d}\boldsymbol{x}$ (integrated over the middle surface Σ of the shell) as a function of time for different values of k_0 and computed from numerical simulations of the radiative transfer equation (41) as described in section 4.4. Here again we considered Gaussian correlations of the perturbations with $\xi_\varrho = 0 \cdot 1$ and $\xi_\chi = \xi_\psi = 1$ for fluctuation amplitudes. We clearly see the convergence of this ratio to its expected limit $\left(\frac{c_S}{c_P}\right)^2 = \frac{1-\nu}{2} = 0 \cdot 4$ for our choice of $\nu = 0 \cdot 2$. The speed of convergence increases as k_0 decreases, in accordance with the evolution of the diffusion constant with this parameter shown in figure 12. Note finally

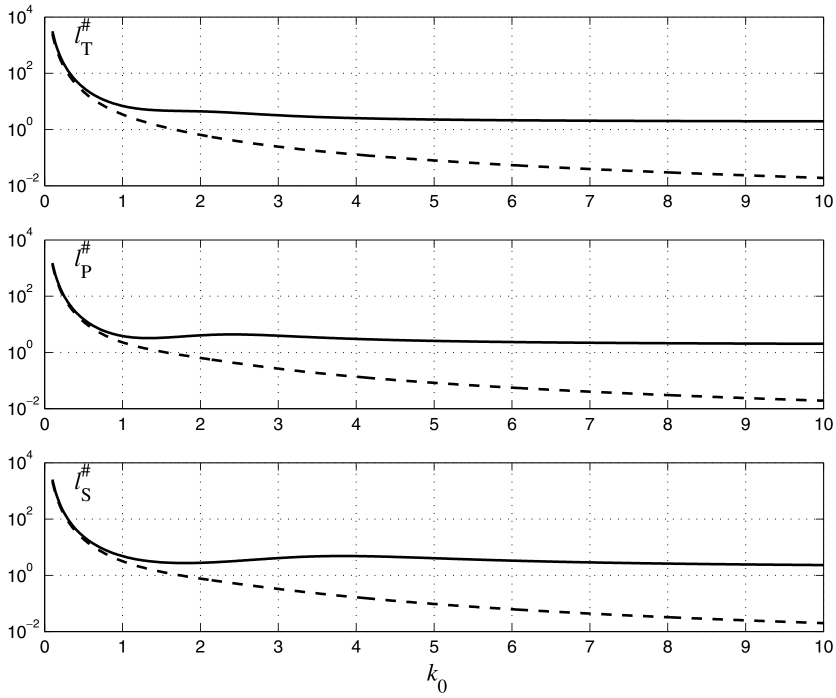

Figure 11. Normalized scattering (- - - -) and transport (——) mean free paths for a thick shell with Gaussian correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$ and $\xi_\chi = \xi_\psi = 1$.

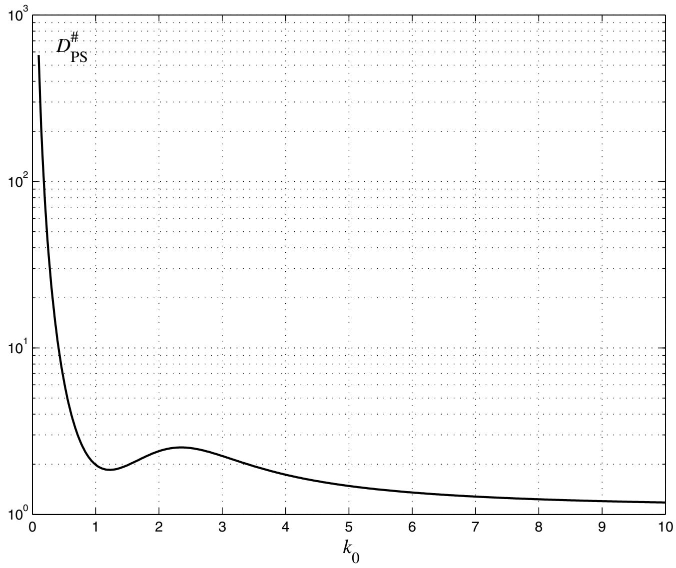

Figure 12. Normalized diffusion coefficient $D^{\#}_{\mathrm{PS}}$ for a thick shell with Gaussian correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$ and $\xi_\chi = \xi_\psi = 1$.

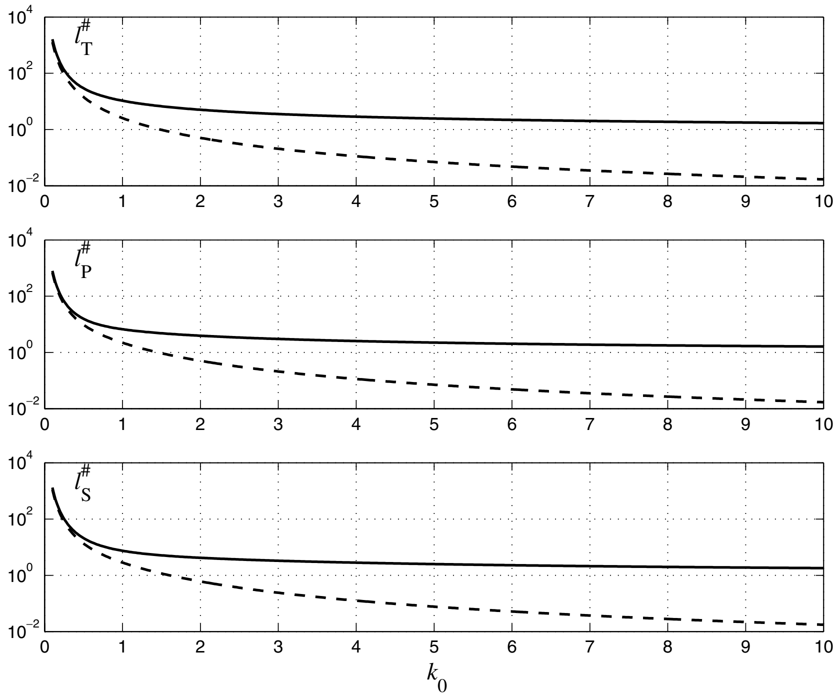

Figure 13. Normalized scattering (- - - -) and transport (——) mean free paths for a thick shell with exponential correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$ and $\xi_\chi = \xi_\psi = 1$.

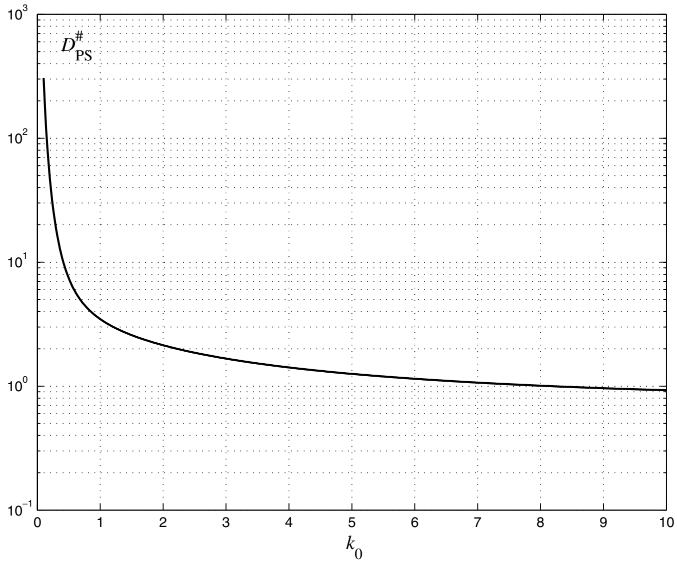

Figure 14. Normalized diffusion coefficient $D_{PS}^\#$ for a thick shell with exponential correlations of the perturbations; $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$ and $\xi_\chi = \xi_\psi = 1$.

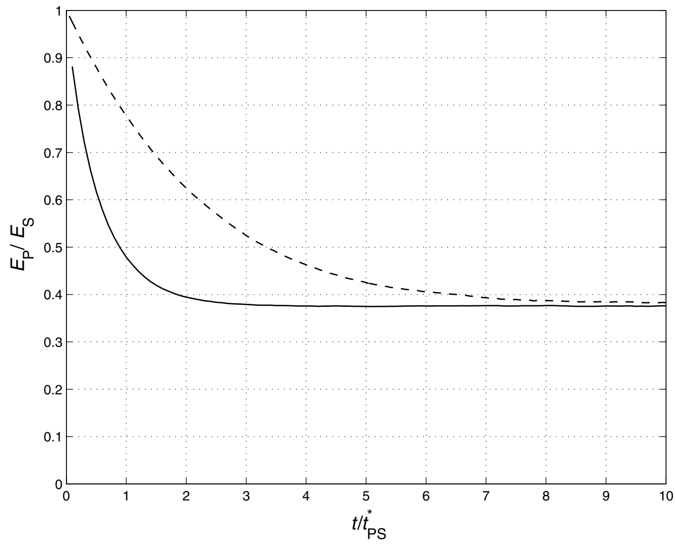

Figure 15. Bending energy ratio $\frac{\mathcal{E}_P}{\mathcal{E}_S}$ as a function of time in a circular cylindrical shell for $k_0 = 1$ (——) and $k_0 = 10$ (- - - -); $\nu = 0{\cdot}2$, $\xi_\varrho = 0{\cdot}1$, $\xi_\chi = \xi_\psi = 1$ and $c_S = 1\,\mathrm{m\,s^{-1}}$.

that as a general rule the exponent in the equipartition law is equal to d ($d = 2$ for a shell), for instance, it is equal to 3 in a three-dimensional elastic medium [23].

6. Conclusions

We have shown in this paper that the high-frequency vibrations of heterogeneous slender structures such as cylindrical shells can be investigated in the frame of the so-called mathematical theory of microlocal analysis. It shows that the energy associated with the high-frequency solutions of elastodynamic equations satisfies transient transport equations, or radiative transfer equations. In the limit of long propagation distances, or short scattering mean free paths, these equations can be approached by diffusion equations. The condition for the emergence of diffusive fields is the presence of many reflectors or scatterers such as heterogeneities in a random medium, or structural components (stiffeners, stringers, panels, boundaries, . . .) with significant stiffness contrasts in a complex structure. The vibrational conductivity approach of high-frequency structural dynamics has already considered a diffusion equation, but the conditions by which it is obtained are not fulfilled in this theory. Furthermore it is limited to the steady-state propagation in one-dimensional wave guides. The proposed theory applies to multi-mode transient propagation in arbitrary three-dimensional structures. It brings a new and highly relevant insight to the problem of the transient dynamic response of elastic structures to impacts or shocks and to the derivation of their diffusion properties. The research is in the progress to modelize rigorous boundary conditions for energy propagation in the bounded media.

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